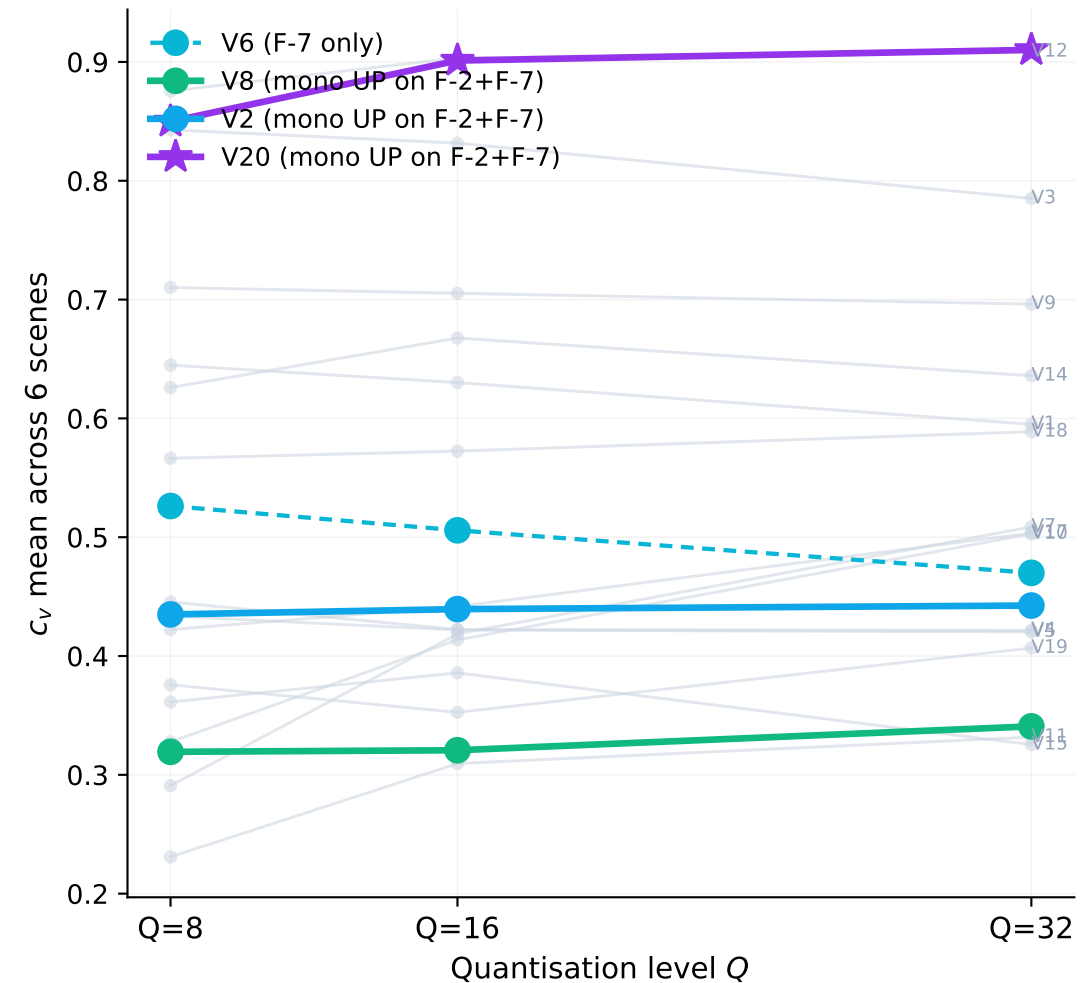
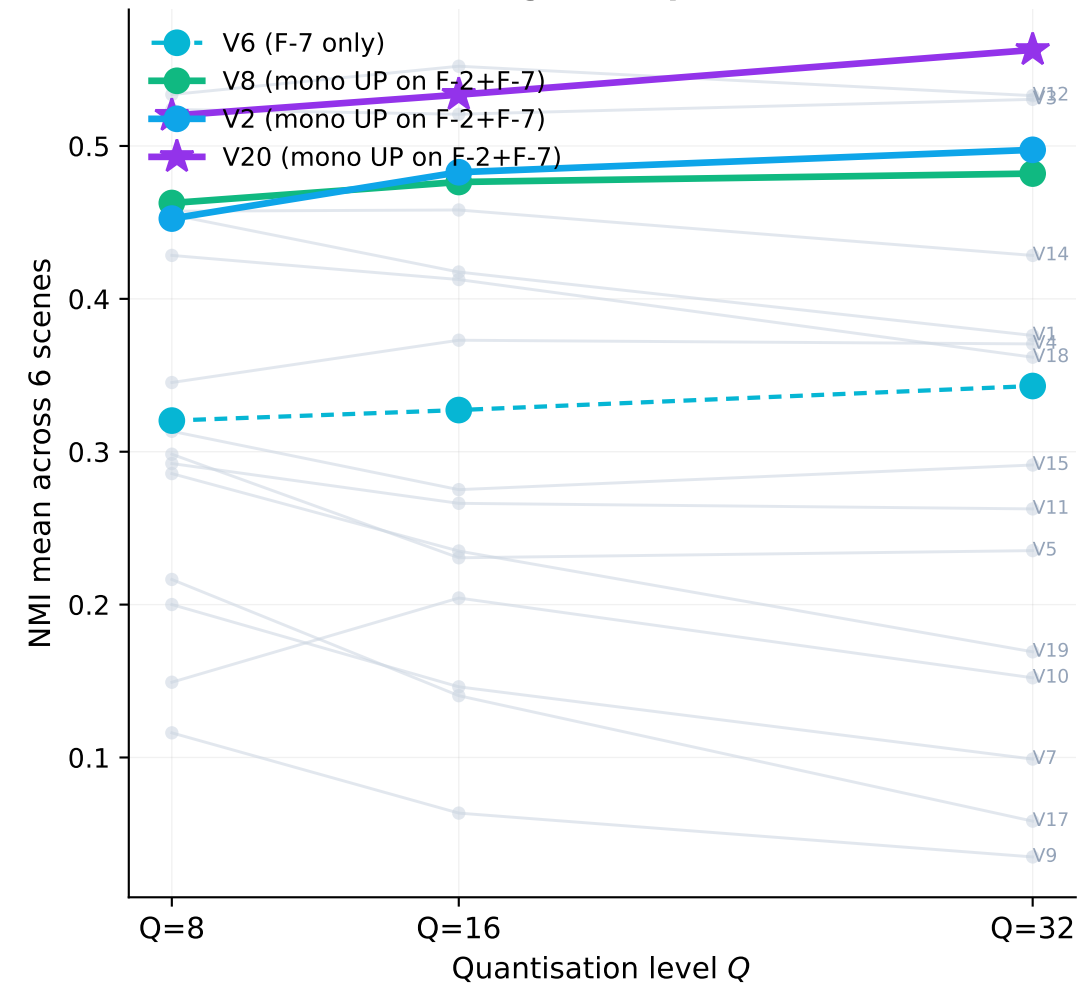


Full 19-recipe Q-sensitivity: only V20/V2/V8 increase monotonically on both axes

F-2 coherence c_v (top-10 words)



F-7 NMI (argmax topic vs label)



Three recipes climb monotonically on F-2 and F-7 across Q=8/16/32: V20 (purple ★, label-aware), V2 (cyan, intensity-bin), V8 (green, NFINDR endmember). V6 (dashed cyan) climbs on F-7 only; its F-2 declines. V20 has the largest F-2 gain of the three and the highest F-7 at Q=32. Of the grey recipes on F-7, seven decline monotonically, four peak at Q=16 and three dip at Q=16.